throat symptoms. "XE is of less concern in a fully vaccinated population," says virologist Dr Gagandeep Kang. According to UKHSA, over 600 cases of XE have been identified till date—less than one per cent of total sequenced strains from Covid cases. Unlike Omicron, which replaced Delta globally in five-six weeks, XE has spread much slower.

That's an encouraging sign, but experts hesitate to rule out the possibility of a future wave, especially once the efficacy of vaccinations and immunity derived from previous infections start to fade. However, experts are hopeful that we will not experience another deadly wave of the type seen in India as the Delta variant spread in 2021.



KNOW YOUR XE

- First known case of the variant was detected in the UK on January 19
- Two suspected cases in India—in Gujarat and Maharashtra
- Evidence suggests XE is 10% more transmissible than other Omicron strains. Till now, 600 cases worldwide
- A 'recombinant' variant, XE emerged when an individual was infected by both BA.1 and BA.2 Omicron strains
- So far, disease caused by XE is assumed to be mild. Unclear how it will impact children, those unvaccinated

"Unlike humans, who have DNA as their genetic material, coronaviruses have RNA, which is less stable and allows them to make new strains and variants in order to survive and replicate. So far, XE hasn't seen rapid spread in the way that BA.2 did. Moreover, with a large percentage of India having been fully vaccinated, the threat of the new variant is less," says Dr Rakesh Mishra, director of the Tata Institute for Genetics and Society. "But Covid restrictions should remain in place as there is a lot left to know about the new variant and there is no guarantee other variants won't come up in future."

While most Indians are fully vaccinated against Covid, questions as to how long their immunity will last and how effective it will be against the XE variant persist. Data from recent surges in the UK and Denmark show that reinfection with XE is possible in as little as 60 days. Reinfection with Omicron by contrast is unlikely to occur within 90 days of a Delta infection. A renewed fortification can, thus, be called for. "Booster doses are important to strengthen immunity at this point," says Dr Farah Ingale, internal medicine head, Fortis Hiranandani, Mumbai.

WHY VACCINES REMAIN VITAL

The Union health ministry from April 10 onwards has announced that all adults can take precautionary doses of vaccines at private centres nine months after their second dose. However, uptake has been slow, with only 9,674 jabs given on day one. An April 2022 survey by LocalCircles showed that 42 per cent of eligible respondents do not want a third jab. Experts say this is largely due to people's perception that Omicron is mild, or that two doses have them well covered.

"Antibody levels fall after some time and getting a booster will help ensure a stronger immune response. Especially since Covid protocols are being relaxed in many states," says Dr Sushila Kataria of Medanta, Gurgaon. Dr Kataria was one of the first to treat Covid patients in Haryana in 2020. "The disease has become milder since then, but this doesn't mean zero impact on health. Those eligible must get vaccinated for their own and the community's sake," Kataria adds. Unlike other countries, India has not allowed different Covid vaccines for the first and second shots. Preliminary clinical trial results reported in *The New England Journal of Medicine*, which served as the basis for recommendations by the US Food and Drug Administration (FDA) and the Centers for Disease Control and Prevention (CDC) to permit mix-and-match booster shots in the US, have shown that different vaccines give a strong immune response.

A study involving 98 people conducted by the Indian Council of Medical Research (ICMR) in Uttar Pradesh's Siddharthnagar district, where 18 people received Covishield as the first dose and Covaxin as the second dose, found that such immunisation was safe and that the antibody response in the mix-and-match group was superior. However, such studies are limited in size and, according to the WHO, there is not enough data on the efficacy of mix-and-match boosters. Even though research has shown that mRNA vaccines—where cells are instructed to make coronavirus proteins that trigger immune response—are more effective as boosters, they are not available in India. Experts also say that getting the same vaccine as a third dose won't reduce its efficacy. "People read things on the internet or on social media and make decisions based on rumour, not medical science. The third dose is important to keep your immunity up," says Aurangabadwala, adding that unrestricted spreading of the virus can spawn more variants, as it has a chance to mutate in every new host. Already, Chinese authorities have identified two new Omicron sub-variants—one derived from BA.1 and one from BA.2—that don't match any existing sequences in China or international databases.

While XE itself is not a cause for concern, experts are worried that by not putting in a concerted global effort to restrict the spread of Covid-19, the opportunity for new variants to come up is growing wider by the day. Their concerns are borne out with Omicron alone reporting two significantly more contagious sub-strains. For the pandemic to subside, accelerated and universal vaccination as well as gimlet-eyed containment are essential. ■